

Some Basic Concepts of Engineering Analysis and an Introduction to the Finite Element Method

3.1 INTRODUCTION

The analysis of an engineering system requires the idealization of the system into a form that can be solved, the formulation of the mathematical model, the solution of this model, and the interpretation of the results (see Section 1.2). The main objective of this chapter is to discuss some classical techniques used for the formulation and solution of mathematical models of engineering systems (see also S. H. Crandall [A]). This discussion will provide a valuable basis for the presentation of finite element procedures in the next chapters. Two categories of mathematical models are considered: lumped-parameter models and continuum-mechanics-based models. We also refer to these as “discrete-system” and “continuous-system” mathematical models.

In a lumped-parameter mathematical model, the actual system response is directly described by the solution of a finite number of state variables. In this chapter we discuss some general procedures that are employed to obtain the governing equations of lumped-parameter models. We consider steady-state, propagation, and eigenvalue problems and also briefly discuss the nature of the solutions of these problems.

For a continuum-mechanics-based mathematical model, the formulation of the governing equations is achieved as for a lumped-parameter model, but instead of a set of algebraic equations for the unknown state variables, differential equations govern the response. The exact solution of the differential equations satisfying all boundary conditions is possible only for relatively simple mathematical models, and numerical procedures must in general be employed. These procedures, in essence, *reduce the continuous-system mathematical model to a discrete idealization* that can be solved in the same manner as a lumped-parameter model. In this chapter we summarize some important classical procedures that are employed to reduce continuous-system mathematical models to lumped-parameter numerical models and briefly show how these classical procedures provide the basis for modern finite element methods.

In practice, the analyst must decide whether an engineering system should be represented by a lumped-parameter or a continuous-system mathematical model and must choose all specifics of the model. Furthermore, if a certain mathematical model is chosen, the analyst must decide how to solve numerically for the response. This is where much of the value of finite element procedures can be found; that is, finite element techniques used in conjunction with the digital computer have enabled the numerical solution of continuous-system mathematical models in a systematic manner and in effect have made possible the practical extension and application of the classical procedures presented in this chapter to very complex engineering systems.

3.2 SOLUTION OF DISCRETE-SYSTEM MATHEMATICAL MODELS

In this section we deal with *discrete* or *lumped-parameter* mathematical models. The essence of a lumped-parameter mathematical model is that the state of the system can be described *directly* with adequate precision by the magnitudes of a finite (and usually small) number of state variables. The solution requires the following steps:

1. *System idealization*: the actual system is idealized as an assemblage of elements
2. *Element equilibrium*: the equilibrium requirements of each element are established in terms of state variables
3. *Element assemblage*: the element interconnection requirements are invoked to establish a set of simultaneous equations for the unknown state variables
4. *Calculation of response*: the simultaneous equations are solved for the state variables, and using the element equilibrium requirements, the response of each element is calculated.

These steps of solution are followed in the analyses of the different types of problems that we consider: steady-state problems, propagation problems, and eigenvalue problems. The objective in this section is to provide an introduction showing how problems in these particular areas are analyzed and to briefly discuss the nature of the solutions. It should be realized that not all types of analysis problems in engineering are considered; however, a large majority of problems do fall naturally into these problem areas. In the examples in this section we consider structural, electrical, fluid flow, and heat transfer problems, and we emphasize that in each of these analyses the same basic steps of solution are followed.

3.2.1 Steady-State Problems

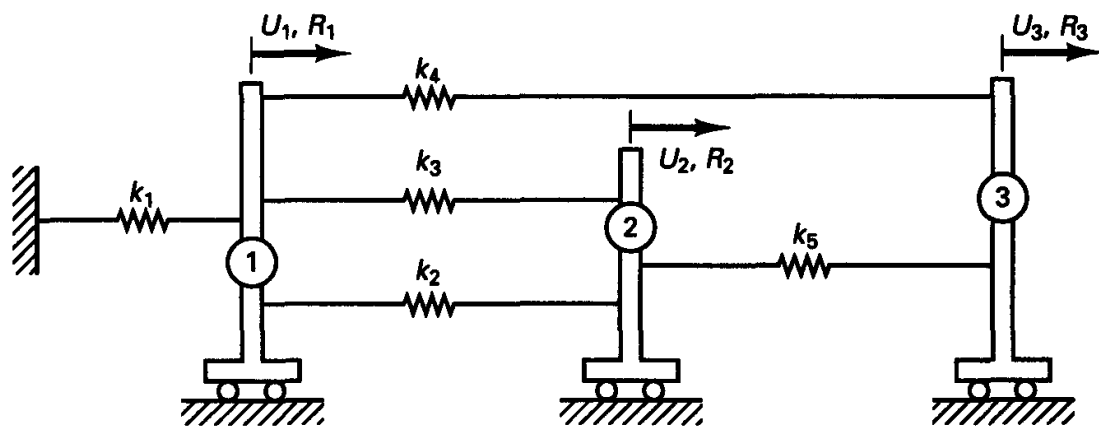
The main characteristic of a steady-state problem is that the response of the system does not change with time. Thus, the state variables describing the response of the system under consideration can be obtained from the solution of a set of equations that do not involve time as a variable. In the following examples we illustrate the procedure of analysis in the solution of some problems. Five sample problems are presented:

1. Elastic spring system
2. Heat transfer system

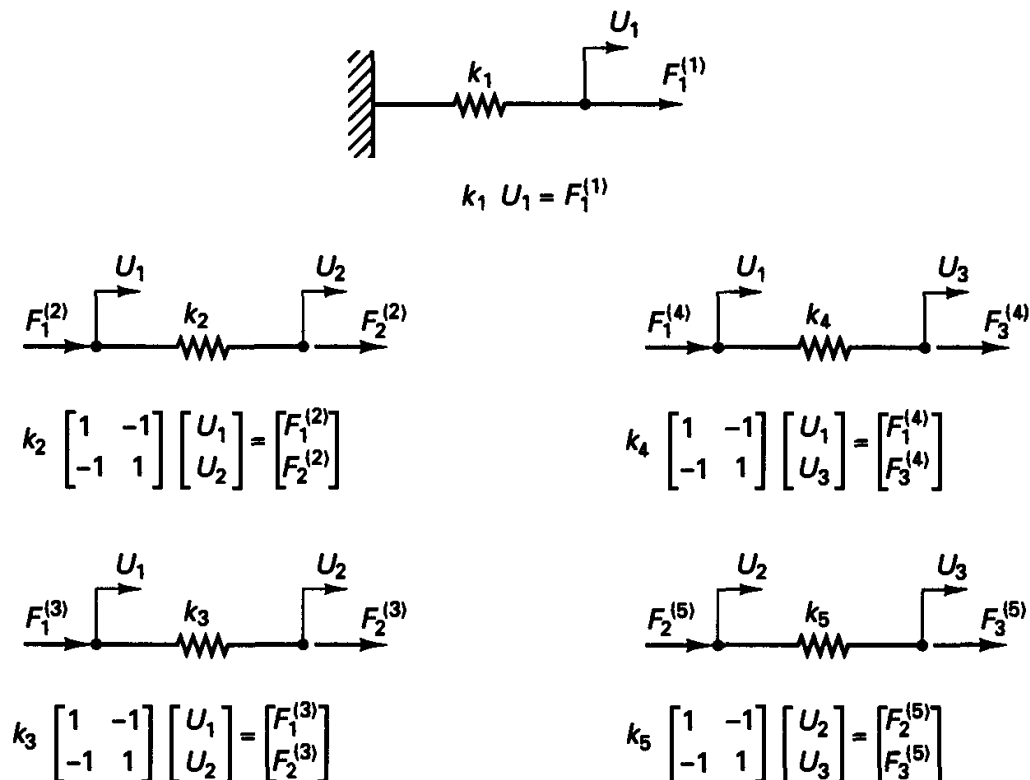
3. Hydraulic network
4. Dc network
5. Nonlinear elastic spring system.

The analysis of each problem illustrates the application of the general steps of analysis summarized in Section 3.2. The first four problems involve the analysis of linear systems, whereas the nonlinear elastic spring system responds nonlinearly to the applied loads. All the problems are well defined, and a unique solution exists for each system response.

EXAMPLE 3.1: Figure E3.1 shows a system of three rigid carts on a horizontal plane that are interconnected by a system of linear elastic springs. Calculate the displacements of the carts and the forces in the springs for the loading shown.



(a) Physical layout



(b) Element equilibrium relations

Figure E3.1 System of rigid carts interconnected by linear springs

We perform the analysis by following steps 1 to 4 in Section 3.2. As state variables that characterize the response of the system, we choose the displacements U_1 , U_2 , and U_3 . These displacements are measured from the initial positions of the carts, in which the springs are unstretched. The individual spring elements and their equilibrium requirements are shown in Fig. E3.1(b).

To generate the governing equations for the state variables we invoke the element interconnection requirements, which correspond to the static equilibrium of the three carts:

$$\begin{aligned} F_1^{(1)} + F_1^{(2)} + F_1^{(3)} + F_1^{(4)} &= R_1 \\ F_2^{(2)} + F_2^{(3)} + F_2^{(5)} &= R_2 \\ F_3^{(4)} + F_3^{(5)} &= R_3 \end{aligned} \quad (a)$$

We can now substitute for the element end forces $F_i^{(j)}$; $i = 1, 2, 3$; $j = 1, \dots, 5$; using the element equilibrium requirements given in Fig. E3.1(b). Here we recognize that corresponding to the displacement components U_1 , U_2 , and U_3 we can write for element 1,

$$\begin{bmatrix} k_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} F_1^{(1)} \\ 0 \\ 0 \end{bmatrix}$$

or

$$\mathbf{K}^{(1)}\mathbf{U} = \mathbf{F}^{(1)}$$

for element 2,

$$\begin{bmatrix} k_2 & -k_2 & 0 \\ -k_2 & k_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} F_1^{(2)} \\ F_2^{(2)} \\ 0 \end{bmatrix}$$

or $\mathbf{K}^{(2)}\mathbf{U} = \mathbf{F}^{(2)}$, and so on. Hence the element interconnection requirements in (a) reduce to

$$\mathbf{KU} = \mathbf{R} \quad (b)$$

where

$$\mathbf{U}^T = [U_1 \quad U_2 \quad U_3]$$

$$\mathbf{K} = \begin{bmatrix} (k_1 + k_2 + k_3 + k_4) & -(k_2 + k_3) & -k_4 \\ -(k_2 + k_3) & (k_2 + k_3 + k_5) & -k_5 \\ -k_4 & -k_5 & (k_4 + k_5) \end{bmatrix}$$

and

$$\mathbf{R}^T = [R_1 \quad R_2 \quad R_3]$$

Here it is noted that the coefficient matrix $\mathbf{K}$ can be obtained using

$$\mathbf{K} = \sum_{i=1}^5 \mathbf{K}^{(i)} \quad (c)$$

where the $\mathbf{K}^{(i)}$ are the element stiffness matrices. The summation process for obtaining in (c) the total structure stiffness matrix by direct summation of the element stiffness matrices is referred to as the *direct stiffness method*.

The analysis of the system is completed by solving (b) for the state variables U_1 , U_2 , and U_3 and then calculating the element forces from the element equilibrium relationships in Fig. E3.1.

EXAMPLE 3.2: A wall is constructed of two homogeneous slabs in contact as shown in Fig. E3.2. In steady-state conditions the temperatures in the wall are characterized by the external surface temperatures θ_1 and θ_3 and the interface temperature θ_2 . Establish the equi-